

Table 3. Representative watermark schemes and fixed operating points used in our experiments. Schemes are selected in advance to cover distinct instantiations within each watermark family, and hyperparameters are held fixed across attacks (i.e., no per-attack retuning).

Family	Method	Representative Role	Fixed Operating Point Used
Biased	KGW (Kirchenbauer et al., 2023)	Canonical context-dependent biased sampling representative	$\gamma = 0.5, \delta = 2.0$, <code>simple-1</code> seeding
Biased	Unigram (Zhao et al., 2024)	Canonical fixed-green-list biased sampling representative	$\gamma = 0.5, \delta = 2.0$
Bias-free	DiPMark (Wu et al., 2024)	Canonical token-level bias-free reweighting representative	$\alpha = 0.45, \gamma = 0.5$
Bias-free	HCW (Hu et al., 2024)	Canonical unbiased reweighting representative distinct from DiPMark	δ -reweight variant (<code>method=delta</code>)
Bias-free	HeavyWater (Tsur et al., 2025)	Recent low-entropy bias-free baseline included to broaden family coverage	Fixed released/reference setting
Bias-free	SimplexWater (Tsur et al., 2025)	Recent low-entropy bias-free baseline included to broaden family coverage	Fixed released/reference setting
Semantic	SemStamp (Hou et al., 2024)	Canonical sentence-level semantic watermark representative	Fixed released/reference setting
Semantic	PMark (Huo et al., 2025)	Canonical rejection-based semantic watermark representative	Rejection parameter $\rho = 0.25$
Semantic	SimMark (Dabiriaghdam & Wang, 2025)	Recent similarity-based semantic baseline included to broaden family coverage	Fixed released/reference setting
Distribution-preserving	CGW (Christ et al., 2024)	Canonical distribution-preserving representative	Security parameter $\lambda = 128$; fixed secret key across runs
Training-time baseline	GaussMark (Block et al., 2025)	Structural baseline reported separately from inference-time families	Gaussian perturbation scale $\sigma = 0.05$ (held-out calibration at $\alpha = 0.05$)

Note. For methods whose released implementations expose additional internal settings beyond the main operating-point parameter (e.g., semantic encoder defaults, detector thresholds, or prompt templates), we follow the released/reference configuration without per-attack retuning.